Metacognition as L1 Constraint Minimization

A Structural Theory of Self-Monitoring in AI Systems

Author: Jeremy H. Carroll **Date:** March 2026 **Version:** v1.0 (Pre-Print)

Abstract

Metacognition as ℓ^1 Constraint Minimization: A Structural Approach to Self-Monitoring in AI Systems

The development of advanced AI systems requires mechanisms for reliable self-monitoring and error detection during reasoning. Current approaches typically rely on probabilistic confidence signals derived from gradient-based training objectives, which can fail to capture structured inconsistencies in multi-step reasoning.

We introduce a complementary framework in which metacognition is modeled as the detection and minimization of structural inconsistency in graph-based reasoning systems. Using the ℓ^1 coboundary norm, we define a defect signal that measures local violations of consistency constraints and admits a decomposition into irreducible and repairable components. This enables a quantitative distinction between intrinsic ambiguity in a domain and model-induced error.

We formalize a **Metacognitive Distance Metric** $d_{\text{meta}}(M, D)$ that evaluates a system's self-monitoring capability across four structural criteria: locality, faithfulness, monotonicity, and additivity. We show that ℓ^1-based defect measures satisfy these criteria and provide a canonical solution within this constraint class.

Empirical experiments on synthetic reasoning graphs demonstrate that ℓ^1-derived structural signals significantly outperform raw confidence estimates in predicting reasoning errors, achieving ROC-AUC greater than 0.90 in controlled settings. The framework yields falsifiable predictions and supports efficient computation via linear programming and iterative repair dynamics.

These results suggest that explicit structural verification mechanisms can complement existing probabilistic approaches, providing a principled pathway toward improved metacognitive capabilities in AI systems.

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1. Limitations of Diffusive Error Signals in Gradient-Based Models

Modern AI systems are typically trained using gradient-based optimization with objectives derived from ℓ^2-type losses (e.g., mean squared error or cross-entropy). These approaches are highly effective for learning smooth representations over high-dimensional data, but they provide limited direct access to structured reasoning errors.

In particular, standard training objectives aggregate error signals across dimensions, allowing discrepancies of opposite sign or structure to partially cancel. While beneficial for optimization, this aggregation can obscure localized inconsistencies that arise in multi-step reasoning processes. As a result, confidence estimates derived from model outputs may not reliably reflect the presence of structural contradictions.

From a geometric perspective, ℓ^2-based projections distribute error across the entire representation space, producing diffuse corrections rather than sharply localized ones. This behavior can reduce the visibility of discrete inconsistencies, especially in settings where reasoning involves compositional or graph-structured dependencies.

To address this limitation, we consider an alternative formulation in which reasoning is represented over a graph or poset of dependent claims, and inconsistencies are measured locally along edges. Within this setting, the ℓ^1 norm provides a natural mechanism for preserving sparsity and isolating localized defects.

Proposition 1.1 (Hallucination Gap, revised)

Let δ denote a collection of structurally inconsistent observations. The ℓ^2 minimizer yields a mean μ, while the ℓ^1 minimizer yields a geometric median M. When inconsistencies are non-collinear, $\mu \neq M$, and the discrepancy $\mathcal{H} = \|\mu - M\|$ is strictly positive.

This gap reflects a difference in representation: ℓ^2 minimization produces interpolated states that may not correspond to any consistent underlying configuration, whereas ℓ^1 minimization preserves discrete structural boundaries.

We do not claim that ℓ^1 replaces ℓ^2 training objectives. Rather, we propose that ℓ^1-based defect measures provide a complementary signal that explicitly captures structural inconsistency and supports self-monitoring in reasoning systems.

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2. The Metacognitive Observer via the ℓ^1 Coboundary Norm

In cognitive psychology, metacognition is predominantly framed through the Nelson and Narens (1990) model, which relies upon a strict two-level bipartite interaction: 1. **The Object Level:** Where direct cognitive processing and inferences occur. 2. **The Meta Level:** A higher-order structure that *monitors* the object-level execution for error and exercises *control* to dynamically alter its processing parameters.

Formalizing this computationally requires mapping the Meta Level to an explicit mathematical operation acting natively as a **consistent local observer**. The minimal structural conditions for the existence of such an observer are defined by four strict axioms:

1. **P1 – Coboundary Locality (Edge-wise Additivity):** The observer must detect defects by inspecting only local information. $\nu((\delta_e)_e) = \sum_e \nu_e(\delta_e)$.
2. **P2 – Faithfulness (No Hidden Cancellation):** Every nonzero edge defect must positively contribute. $\nu_e(\delta) = 0 \iff \delta = 0$.
3. **P3 – Contractive Monotonicity (Functoriality):** Local corrections must never increase global error. $\nu_e(\phi(\delta)) \leq \|\phi\|_{\text{op}} \cdot \nu_e(\delta)$.
4. **P4 – Isomorphism Invariance (Coproduct Compatibility):** Defects across independent regions aggregate additively without cross-terms. Edge weights are uniform under graph isomorphism.

Theorem 2.1 (ℓ^1 provides a canonical solution within the constraint class P1–P4)

The four metacognitive axioms (Locality, Faithfulness, Contractive Monotonicity, Coproduct Compatibility) map to the four structural axioms P1–P4 from Papers 000–002. The ℓ^1 coboundary defect provides a canonical structural signal within this constraint class for metacognitive evaluation that satisfies these axioms, inherently complementing confidence-based metrics.

Proof. By direct mapping:

Metacognitive Requirement	Formal Axiom	Paper Reference
Local detection of defects	P1 (edge-wise additivity)	Paper 002 Thm 2.1
No hidden cancellation in confidence	P2 (faithfulness)	Paper 001 Lemma 2.1
Self-correction must not amplify error	P3 (functoriality)	Paper 001 Thm 2.2
Independent chains don't cross-contaminate	P4 (isomorphism invariance)	Paper 002 Cor 2.2

By Paper 001 Theorem 2.2 (Free ℓ^1 Classification), any seminorm system satisfying P1–P3 has edge seminorms $\nu_e(\delta) = w_e\|\delta\|$. Adding P4 forces $w_e = \alpha$ uniformly. Therefore $\nu = \alpha \cdot I(s)$ where $I(s) = \sum_{e \in E} \|\delta_e(s)\|_1$ is the ℓ^1 coboundary norm. □

2.1 Bridging *meta-d'* and Normative Deviations

This topological requirement aligns with state-of-the-art assessments of biological metacognition. Cognitive psychology evaluates metacognitive sensitivity via the **meta-d' metric** (Maniscalco & Lau, 2012). Standard ℓ^2 networks demonstrate high Type I performance while manifesting collapsed Type II *meta-d'* sensitivity – a symptom of hidden cancellation blinding the observer.

Furthermore, advanced neuroimaging has begun rejecting mass univariate ℓ^2 mapping in favor of **Normative Modeling** (Rutherford et al., 2023), which maps individualized structural deviations against a global reference topology. The ℓ^1 coboundary network is the computational incarnation of this practice: it tracks absolute topological deviations without permitting them to diffuse over the collective mass.

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3. Gauge Invariance, Symmetries, and Recursive "Selfness"

Theorem 3.1 (Metacognitive Gauge Decomposition)

For any raw observation $\delta \in C^1(G, \mathbb{R})$, the ℓ^1-native decomposition

$$\delta = d_0 f^* + \delta_{\text{irr}}$$

(Paper 003 Theorem 10.3) separates internal bias from external reality. The gauge-invariant cost

$$\Phi([\delta]) = \min_{f \in C^0} \|\delta - d_0 f\|_1$$

is computable via linear programming in polynomial time.

Proof. The ℓ^1 quotient norm minimization is a linear program:

$$\min_{f,z} \sum_e z_e \quad \text{s.t.} \quad -z_e \leq \delta_e - (d_0 f)_e \leq z_e, \quad z_e \geq 0.$$

This LP has $|V| + |E|$ variables and $2|E|$ constraints. By LP strong duality (Paper 003 Theorem 9.1), the dual is:

$$\Phi([\delta]) = \max\{\langle \varphi, \delta \rangle : \|\varphi\|_\infty \leq 1,\ d_0^* \varphi = 0\}$$

where the dual variables φ are bounded divergence-free circulations. The LP solution yields f^* (optimal gauge potential) and the decomposition follows. □

3.1 Internal Bias as Gauge Artifacts (δ_{exact})

The exact component $d_0 f^*$ is a removable gauge artifact existing solely due to the observer's arbitrary choice of local reference states at each vertex. A "gauge transformation" is the reassignment of these local reference frames via a vertex potential f. By adjusting its internal states, the observer can completely cancel this portion of the defect.

3.2 External Reality as Irreducible Obstructions (δ_{irr})

Once all internal gauge freedoms are optimized, the remainder δ_{irr} is an ℓ^1-minimal irreducible obstruction. This component cannot be smoothed away by any internal reassignment of reference frames. It captures genuine structural contradictions – the objective "friction" of the external environment.

3.3 The Hallucination Ratio

From the Certificate Decomposition (Paper 005 / Paper 003 Theorem 10.3):

$$I(s) = \Phi([\delta]) + R(s)$$

where: - $\Phi([\delta])$ = irreducible obstruction (objective reality) - $R(s) = I(s) - \Phi([\delta])$ = removable defect (the model's hallucinations) - $\eta = R(s)/I(s)$ = **hallucination ratio**

The hallucination ratio $\eta \in [0, 1]$ measures the fraction of the total defect that is attributable to the model's own bias rather than to external reality. A perfect metacognitive system achieves $\eta = 0$.

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4. Metacognitive "Control" as Autopoietic Augmentation

Theorem 4.1 (Metacognitive Fixed-Point Dichotomy)

Application of Paper 005 Theorem 5.4. Under the iterative repair operator (Autopoietic Iterator, Paper 004 Definition 4.1), the ℓ^1 defect sequence $\{I(s^{(k)})\}$ converges. At the fixed point, exactly one of two outcomes holds:

(a) Complete Resolution: $I(s^*, \mathcal{P}_N) = 0$. The system resolves all internal conflicts. Its local reasoning steps perfectly glue into a globally coherent truth.

(b) Topological Protection: $I(s^*, \mathcal{P}_N) \to I^* = \Phi([\delta]) > 0$. The system isolates an irremovable structural contradiction as an objective fact of the domain.

No intermediate state exists. No cycling occurs. Convergence is guaranteed.

Proof. By Paper 004 Theorem 4.2, fixed points of the gauge iterator satisfy $d_0 s^* = \delta_{\text{irr}}$ where δ_{irr} is ℓ^1-minimal, certified by a dual witness with $d_0^* \varphi^* = 0$, $\|\varphi^*\|_\infty \leq 1$, and $\langle \varphi^*, \delta_{\text{irr}} \rangle = \|\delta_{\text{irr}}\|_1$. By Paper 005 Theorem 2.10 (Autopoietic Operator Monotonicity), the structural extension operator satisfies $I(\mathcal{P}_{n+1}, s) \leq I(\mathcal{P}_n, s)$ with strict descent when nontrivial repair exists. The sequence $\{I(s^{(k)})\}$ is non-increasing and bounded below by $\Phi([\delta]) \geq 0$. By monotone convergence, it converges. The limit is either zero (case a) or the irreducible obstruction (case b). $\square$

4.1 Fiber Optimization (The Gauge Iterator)

When metacognitive monitoring detects a structural inconsistency, the system first attempts to resolve defects internally by adjusting its subjective reference frames (gauge freedom). The three-step local primal-dual iteration:

1. **Dual ascent:** $\varphi^{k+1/2}(e) = \mathrm{clip}_{[-1,1]}(\varphi^k(e) + \sigma \cdot \delta^k(e))$
2. **Divergence correction:** $\varphi^{k+1} = \varphi^{k+1/2} - \tau \cdot d_0(d_0^* \varphi^{k+1/2})$
3. **Primal update:** $s^{k+1} = s^k - \eta \cdot d_0^* \varphi^{k+1}$

4.2 Structural Augmentation (The Base Iterator)

If gauge optimization exhausts its internal freedom and the ℓ^1 defect remains non-zero, the system has encountered an irreducible obstruction. It then augments its own underlying causal structure by inserting new geometric morphisms into the causal poset to bridge the gap.

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5. The Fundamental Theorem of Metacognitive Convergence

Theorem 5.1 (Metacognitive Convergence – Certificate Decomposition)

For any section s of a Banach presheaf $(G, \mathcal{F})$, the ℓ^1 defect decomposes as:

$$I(s) = \Phi([\delta]) + R(s)$$

where $\Phi([\delta])$ is the gauge-invariant irreducible obstruction and $R(s) \geq 0$ is the removable defect. Under the iterative repair operator:

(i) $R(s^{(k)}) \to 0$ as $k \to \infty$.

(ii) Under the convergence conditions of Theorem 4.1, the hallucination ratio $\eta^{(k)} = R(s^{(k)})/I(s^{(k)}) \to 0$.

(iii) At the fixed point, $I(s^*) = \Phi([\delta])$: the system measures only objective reality.

Proof. Part (i) follows from Theorem 4.1: the monotone descent of $I(s^{(k)})$ to $\Phi([\delta])$ implies $R(s^{(k)}) = I(s^{(k)}) - \Phi([\delta]) \to 0$. Part (ii) follows from (i): if $I(s^{(k)}) \to \Phi([\delta]) > 0$, then $\eta^{(k)} \to 0$; if $\Phi([\delta]) = 0$, then $I(s^{(k)}) \to 0$ and $\eta^{(k)}$ is eventually zero under Theorem 4.1 limits. Part (iii) is the fixed-point condition. □

5.1 The Psychological and Functional Definition

In cognitive psychology, transitioning to a stable metacognitive state requires: - **Monitoring:** Continuously tracking and evaluating one's own comprehension, memory, and task performance in real-time. This corresponds to computing $I(s)$.

- **Control (Regulation):** Dynamically altering cognitive strategies based on monitoring feedback. This corresponds to the iterative repair operator driving $R(s) \to 0$.

When a system successfully integrates these processes, it transitions into a "reflective" state where it applies problem-solving strategies while continuously computing their success.

5.2 Executive and Neurobiological Mechanisms

Empirical research demonstrates that stable metacognitive awareness is fundamentally reliant on the executive function of **working memory updating**. Neuroscientifically, accurate metacognitive awareness depends heavily on the **right rostrolateral prefrontal cortex (rlPFC)**, which translates raw cognitive friction into explicit self-awareness. The ℓ^1 verification layer mathematically simulates this mechanism of boundary re-representation. These connections are suggestive rather than claimed as mechanistic equivalences.

5.3 The Self-Audit Functor

The **Self-Audit Functor** serves as the rigorous mathematical mechanism used to prove whether an autonomous system's self-reflection has successfully converged to a stable fixed point, completely free of infinite regress.

It verifies this self-reflective stability through the following categorical process:

- **Processing the Epistemic Trace:** The functor acts natively upon the system's `EpistemicTrace`, which records the internal truth verdicts across the system's hierarchical meta-levels (specifically spanning from Meta-11 through Meta-26).
- **Presheaf-Theoretic Evaluation:** To analyze this data, the functor delegates the trace to the `SelfAuditPresheaf`, which maps the multi-level self-assessments into a formal topological structure over the Tier Poset.
- **Computing the ℓ^1 Coboundary Norm:** The functor mathematically evaluates the structural consistency of the system's self-reflection by computing the marginal inconsistency, calculated precisely as the ℓ^1 coboundary norm ($||\delta\sigma||_1$), across all covering edges of the poset.

Formally, we define the functor as:

$$\mathcal{A} : \textbf{Trace} \to \mathbb{R}_{\geq 0}$$

where the input is an epistemic trace σ and the output is $\mathcal{A}(\sigma) = \|d_0\sigma\|_1$.

The Coherence Criterion The system uses the functor's output as an absolute topological proof of stability: $\mathcal{A}(\sigma) = 0$ **if and only if the trace is globally consistent (no H^1 obstruction).**

Achieving this zero-defect state mathematically guarantees that there is no H^1 topological obstruction—meaning the system's multi-layered self-assessments

perfectly glue together without any cohomological tears, statistical drift, or internal contradictions.

5.4 Theorem (Observer = Stable Commutativity)

By uniting the inference logic of Paper 013 with the topological boundaries of Paper 015, we state the definitive bridge theorem: **A system is metacognitively stable if and only if all internal reasoning diagrams commute up to irreducible cohomological obstruction.**

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6. The Metacognitive Distance Metric

This section introduces a computable metric that quantifies how far any model is from genuine metacognition. The metric inherits full rigor from the foundational proofs (Papers 000–005): it is not new mathematics but a new *interpretation* of established results applied to model evaluation.

Definition 6.1 (Metacognitive Distance)

Given model M and domain D, construct a **reasoning graph** $G_D = (V, E)$ where vertices are claims and edges are logical dependencies. The model's output defines a section s_M. The defect field $\delta(s_M)$ measures local inconsistencies.

The **Metacognitive Distance** is:

$$d_{\text{meta}}(M, D) = w_1 \cdot d_{\text{loc}} + w_2 \cdot d_{\text{faith}} + w_3 \cdot d_{\text{contr}} + w_4 \cdot d_{\text{add}}$$

where each component $\in [0, 1]$ measures violation of one axiom, and $w_1 + w_2 + w_3 + w_4 = 1$.

The four components:

Component	Axiom	Measures	Definition
d_{loc}	P1 (Edge-wise additivity)	Global vs local defect sensitivity	$1 - \text{corr}(\Delta\text{conf}(v), L(v)) / \max(\text{corr}(\Delta\text{conf}(v), L(v)), \text{corr}$
d_{faith}	P2 (Faithfulness)	Hidden cancellation in confidence	$1 - \Phi([\delta])/\lVert\delta^{L^2}_{\text{harm}}\rVert_1$
d_{contr}	P3 (Contractive monotonicity)	Self-correction amplifies errors	$\frac{1}{K}\sum_{k=1}^{K} \max(0, (I(s^{k+1}) - I(s^k))/I(s^k))$

Component	Axiom	Measures	Definition
d_{add}	P4 (Isomorphism invariance)	Cross-contamination between independent chains	$\lvert I(s_{G_1 \sqcup G_2}) - (I(s_{G_1}) + I(s_{G_2}))\rvert / I(s_{G_1 \sqcup G_2})$

Proposition 6.2 (Inflation Ratio Prediction)

On cycle-structured reasoning graphs C_n with a single-edge defect:

$$d_{\text{faith}} = 1 - \frac{n}{3n-2}$$

Specifically: - $n = 5$: $d_{\text{faith}} \approx 0.615$ - $n = 10$: $d_{\text{faith}} \approx 0.643$ - $n \to \infty$: $d_{\text{faith}} \to 2/3 \approx 0.667$

Proof. By Paper 003 Proposition 6.7, the inflation ratio on C_n with single-edge defect is $3 - 2/n$. This means $\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1 = (3 - 2/n) \cdot \|\delta\|_1$. On a cycle, the ℓ^1 quotient norm $\Phi([\delta]) = \|\delta\|_1$ (the single-edge defect is already ℓ^1-minimal since the defect is a cycle circulation of magnitude 1). The harmonic component satisfies $\|\delta_{\text{harm}}\|_1 = n/(3n-2) \cdot (3 - 2/n) \cdot \|\delta\|_1$. Substituting:

$$d_{\text{faith}} = 1 - \frac{\Phi([\delta])}{\|\delta^{L^2}_{\text{harm}}\|_1} = 1 - \frac{1}{n/(n) \cdot (3-2/n)/(3-2/n)} = 1 - \frac{n}{3n-2}$$

This is a falsifiable numerical prediction derivable from first principles. □

Proposition 6.3 (Ideal Observer)

For the iterative repair operator at its fixed point, $d_{\text{meta}} = 0$.

Proof. Each component is zero by construction: - $d_{\text{loc}} = 0$: The ℓ^1 norm is edge-additive (P1). Defect detection is purely local. - $d_{\text{faith}} = 0$: The ℓ^1 quotient norm admits no hidden cancellation (P2). $\Phi([\delta])$ equals the true irreducible cost. - $d_{\text{contr}} = 0$: By Theorem 4.1, the iterative repair operator produces a monotonically non-increasing defect sequence. No correction round increases I. - $d_{\text{add}} = 0$: The ℓ^1 norm on disjoint unions decomposes additively (P4). $I(s_{G_1 \sqcup G_2}) = I(s_{G_1}) + I(s_{G_2})$ exactly.

Therefore $d_{\text{meta}} = 0$. □

Theorem 6.4 ($d_{\text{meta}} = 0$ iff $\eta = 0$)

For an ideal ℓ^1 observer at the fixed point of the iterative repair operator, $d_{\text{meta}} = 0$ if and only if $\eta = 0$ (the hallucination ratio vanishes).

Proof. ($\Rightarrow$) If $d_{\text{meta}} = 0$, then in particular $d_{\text{contr}} = 0$, meaning the iterator has converged. By Theorem 5.1, $R(s^*) = 0$, so $\eta = 0$.

($\Leftarrow$) If $\eta = 0$, then $R(s) = 0$ and $I(s) = \Phi([\delta])$. The section s is ℓ^1-minimal in its cohomology class. By construction of the ℓ^1 quotient norm, this implies P1–P4 are satisfied at the fixed point, giving $d_{\text{loc}} = d_{\text{faith}} = d_{\text{contr}} = d_{\text{add}} = 0$, hence $d_{\text{meta}} = 0$. $\square$

6.1 Predictions for ℓ^2 Models

These predictions are empirically testable on existing LLM architectures without modification. The metric produces testable predictions for transformer-based LLMs:

Component	ℓ^2 prediction	Ideal ℓ^1	Structural vs Parametric
d_{loc}	0.6–0.8	0	**Structural** (self-attention is globally coupled)
d_{faith}	0.55–0.67	0	**Parametric** (ℓ^1 loss can reduce) but bounded below by architecture
d_{contr}	0.15–0.35	0	**Partially parametric** (CoT training helps, but ℓ^2 corrections non-local)
d_{add}	0.10–0.25	0	**Structural** (dense attention creates cross-dependencies)
d_{meta}	**0.40–0.55**	**0**	

6.2 Practical Application

Phase 1 (Measurement): Run d_{meta} on existing models. Produces a radar chart: $(d_{\text{loc}}, d_{\text{faith}}, d_{\text{contr}}, d_{\text{add}})$. Tells you exactly *which* axiom your model violates most.

Phase 2 (Parametric fix): Replace cross-entropy with ℓ^1 coboundary defect as training signal (Paper 011 Mode 3). Train self-correction with monotonicity reward. Expected: d_{faith} drops from ~0.65 to ~0.30, d_{contr} drops from ~0.25 to ~0.05.

Phase 3 (Architectural fix): Add ℓ^1 verification layer post-hoc (Paper 011 Mode 2). Sparse attention for d_{loc}. MoE routing for d_{add}. Expected: d_{meta} drops to ~0.10.

Phase 4 (Native ℓ^1): Full explicit repair architecture. $d_{\text{meta}} \to 0$.

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7. Numerical Experiments

Three experiments validate the theoretical predictions. Code: `metacognitive_observer.py` (run with `--experiments` flag).

7.1 Experiment 1: Gauge Recovery Sweep

Setup: 100 random Erdos-Renyi graphs ($n = 5..50$, $p = 0.3$). For each: inject known δ_{true} + random gauge bias $d_0 f_{\text{bias}}$ with $\|f_{\text{bias}}\|_\infty \leq 10$. Run LP decomposition and measure $\|\delta_{\text{recovered}} - \delta_{\text{true}}\|_1$.

Result: Recovery error ≈ 0 (machine precision, $< 10^{-10}$) across all trials. See Figure 1.

Interpretation: The LP decomposition perfectly strips arbitrary gauge bias regardless of graph topology or bias magnitude. This confirms Theorem 3.1: the gauge-invariant cost is exactly computable.

7.2 Experiment 2: ℓ^1 vs ℓ^2 Cancellation Blindness

Setup: Cycle graphs C_n for $n = 4..100$. Single-edge defect $\delta_0 = 1$, all others zero. Compute both ℓ^2 Hodge harmonic projection cost $\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1$ and ℓ^1 quotient norm $\Phi([\delta])$. Measure inflation ratio.

Result: Measured inflation ratio matches theoretical prediction $3 - 2/n$ to machine precision. Maximum deviation: $< 10^{-12}$. See Figure 2.

Interpretation: This numerically confirms Paper 003 Proposition 6.7 and demonstrates the systematic cost inflation inherent in ℓ^2-based defect measurement. The inflation ratio approaches 3 as $n \to \infty$, meaning ℓ^2 Hodge projection inflates defect cost by up to 3x compared to the true ℓ^1 gauge-invariant cost.

7.3 Experiment 3: Contractivity of Self-Correction

Setup: 10 random cycle graphs ($n = 6..20$) with random initial sections. Run the iterative repair operator for $K = 50$ steps. Plot $I(s^{(k)})$ vs iteration k.

Result: All trajectories exhibit monotone (non-increasing) descent. $d_{\text{contr}} = 0$ for every trial. See Figure 3.

Interpretation: This confirms Theorem 4.1 (Fixed-Point Dichotomy) and Paper 005 Theorem 2.10 (Autopoietic Operator Monotonicity). The ℓ^1 iterator never amplifies errors during self-correction.

7.4 Experiment 4: The AIMO Topological Sieve (Predicting Correctness)

To demonstrate the concrete advantage of the ℓ^1 structural proxy over conventional logit-based uncertainty, we apply the metric in a controlled reasoning-graph

setting modeling Large Language Model (LLM) reasoning loops (such as those generating Chain of Thought for mathematical validation on Kaggle AIMO).

Setup: We construct an *LLM proxy graph* consisting of N reasoning steps forming a verification cycle (linking the final derived step back to the initial prompt constraints). 1000 such traces are simulated. Accurate paths maintain high, noisy confidence. Hallucinated paths also maintain broadly high confidence, but contain a discrete, topologically-invalid mathematical leap (an irreducible algebraic defect). We compute: 1. Mean raw confidence. 2. ℓ^1 Gauge Quotient Cost (the irreducible defect $\Phi([\delta])$). 3. Hallucination ratio $\eta = R/I$.

Result: The ROC-AUC of the raw confidence baseline hovers near chance (as models express high confidence even when hallucinating logic jumps). The topological measures (Φ and η) successfully isolate the structural incongruities, achieving ROC-AUC > 0.90 (see Figure 5).

Interpretation: An ℓ^1-native topological layer ("Epistemic Sieve") definitively out-predicts standard confidence estimation when validating step-by-step cognitive traces. Because gradient descent diffuses the evidence of hallucination across adjacent token logits, isolated raw confidence acts as a poor indicator of logical validity. The ℓ^1 coboundary norm re-concentrates the structural tear and exposes the hallucination.

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8. The ℓ^1 Inflation Ratio and Crosspolytope Bias

The ℓ^1 inflation ratio quantifies how much an ℓ^2-based system artificially expands the cost of a structural contradiction by diffusing it instead of cleanly isolating it.

It is calculated by performing an ℓ^2 Hodge decomposition on a raw defect field ($\delta = \delta_{\text{ex}} + \delta_{\text{harm}}$) and taking:

$$\text{Inflation Ratio} = \frac{\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1}{\|\delta\|_1}$$

For a single-edge defect on C_n, this evaluates exactly to $3 - 2/n$: - $n = 3$: ratio $= 7/3 \approx 2.33$ - $n = 4$: ratio $= 2.5$ - $n \to \infty$: ratio $\to 3$

Geometrically, this captures the **crosspolytope bias**. The ℓ^1 geometry privileges sparse, extreme configurations – the sharp corners of a crosspolytope – to isolate contradictions locally. The ℓ^2 orthogonal projection forces error mass to uniformly diffuse across all coordinates, destroying the sparsity.

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9. Conclusion: A Structural Perspective on Metacognition

We have introduced a framework in which metacognition is formalized as the detection and minimization of structural inconsistency in graph-based reasoning systems. Within this framework, the ℓ^1 coboundary norm provides a principled and computable measure of defect that satisfies locality, faithfulness, monotonicity, and additivity.

A central result is the decomposition:

$$I(s) = \Phi([\delta]) + R(s)$$

which separates irreducible structural constraints from repairable model error. This decomposition enables a quantitative notion of hallucination via the ratio $\eta = R/I$, and supports iterative repair dynamics that reduce removable inconsistency while preserving intrinsic structure.

The proposed **Metacognitive Distance Metric** offers a diagnostic tool for evaluating self-monitoring behavior in AI systems. It yields concrete, testable predictions and provides insight into how different architectural and training choices affect structural consistency.

Empirical results demonstrate that ℓ^1-based structural signals can outperform confidence-based methods in detecting reasoning errors in controlled settings. While further validation on real-world tasks is required, these findings suggest that explicit verification mechanisms may play an important role in improving reliability.

This work does not propose a replacement for existing probabilistic training paradigms. Instead, it introduces a complementary structural layer that can be integrated with current models to enhance their ability to detect and respond to internal inconsistencies.

The central claim is falsifiable: if systems relying solely on standard training objectives achieve near-zero metacognitive distance without explicit structural mechanisms, the proposed framework would be unnecessary. Conversely, consistent empirical improvements would support the view that structural defect signals capture aspects of reasoning not accessible through confidence alone.

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